02 · Does the offer work better for some segments? — segment effects (pathmc)

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The business decision. The email lifts spend *on average* (that's the ATE from notebook 01). But an average can hide everything that matters: maybe the email really moves **high-value, highly-engaged** customers and barely touches the rest. If the segments genuinely differ, we should write segment-specific send rules; if the apparent "gap" is just noise, chasing it would be overfitting our marketing to randomness. Either way we want to know **what it costs to treat everyone as the average**.

1.0.1 The concept: moderation and the CATE

When a treatment's effect *depends on who receives it*, we say the effect is **moderated** by the customer's characteristics, and the characteristic (here, being high-value, or an engagement score) is a **moderator**. Formally we're estimating the **CATE** — the *Conditional* Average Treatment Effect, $\tau(x) = \mathbb{E}[Y(1) - Y(0) \mid X = x]$: the average email effect *among customers who look like* x. ($Y(1), Y(0)$ are the two potential outcomes — spend if emailed / not — from notebook 01; $\mathbb{E}[\cdot]$ is "average over customers.") The ATE is just the CATE averaged over everybody.

1.0.2 The tool: a structural model + the do-operator

Notebook 01 used flexible trees (BART) to get the CATE. Here we use a different, more *interpretable* approach — a **structural causal model** via the `pathmc` library. We write down the equation we believe generated spend (including an **interaction term** — a feature that multiplies the treatment, which is exactly how "the effect depends on a moderator" is encoded), fit it, and then ask the model counterfactual questions with the **do-operator**: `do(email=1)` means "*intervene* to email this customer," as opposed to merely *observing* emailed customers (who differ in other ways). The difference in predicted spend between `do(email=1)` and `do(email=0)`, at a given x, *is* the CATE at x.

The single most important output is the **posterior of the interaction coefficient**: if it's clearly away from zero, the segments *really* respond differently; if it straddles zero, we can't tell the gap from noise — so we shouldn't build segment rules on it (and might instead run a bigger test). That's the "is the difference real?" test, answered with uncertainty.

7-step contract: question -> simulate -> identify -> estimate -> validate -> decide in € -> caveats.

1.1 2 · Simulate a ground truth

Customers are **high- or low-value** (`prior_value`) and also carry a continuous **`engagement`** score. We plant a real, layered moderation: the email lifts low-value customers by **€3** and high-value by **€12** *at zero engagement*, **plus** a smooth **+€8 x engagement** slope on top — so at the **average** engagement (0.5) the segment effects work out to **€7** (low) and **€16** (high), which is what the recovery below is graded against. The true per-customer effect is `tau = 3 + 9·is_high + 8·engagement` — a genuine, sizeable, and *continuous* heterogeneity, spanning €3 (low value, zero engagement) to ~€20 (high value, full engagement). `prior_value` also drives spend directly (must be included); there's a background `trend`.

> **On real data.** There's no single public dataset for this exact setup, but the swap is trivial: your **own CRM / campaign log** already has a segment column (`prior_value`), covariates, a treatment flag (was the customer emailed) and an outcome (spend). Point the model at those columns instead of the simulator. The one requirement is that the email was assigned randomly *or* that you've measured the confounders (notebook 05) — otherwise the "segment effects" are contaminated by who-got-emailed.

```
TRUE segment-average effects (at mean engagement): {'low': 7.0, 'high': 16.0}
TRUE effect spans €3.0 … €19.9 across customers
```

	email	prior_value	engagement	trend	spend	is_high
0	0.0	low	0.918802	58.466377	23.337758	0.0
1	0.0	low	0.651170	60.223782	19.057907	0.0
2	1.0	high	0.242720	53.530129	50.963256	1.0
3	0.0	low	0.119019	37.977148	12.868787	0.0
4	1.0	low	0.488445	61.807028	29.494100	0.0

1.2 3 · Identify — the CATE as a functional of the model

Before fitting anything, we pin down *exactly* what we're estimating and whether it's recoverable.

The estimand. The conditional effect, written with the do-operator (intervene, don't just observe):

$$\tau(x) = \mathbb{E}[\text{spend} \mid do(\text{email}{=}1),\, x] - \mathbb{E}[\text{spend} \mid do(\text{email}{=}0),\, x].$$

Where heterogeneity lives. We assume spend is generated by the structural equation (with e=email, v=is_high, g=engagement)

$$\text{spend} = \beta_0 + \beta_1 e + \beta_2 v + \beta_3\,(e{\cdot}v) + \beta_4\,(e{\cdot}g) + \ldots$$

Take the difference between e=1 and e=0 and everything not multiplied by e cancels, leaving

$$\tau(x) = \beta_1 + \beta_3\, v + \beta_4\, g.$$

So the email effect is a **baseline** β_1 **plus moderation terms**: **all the heterogeneity is carried by the interaction coefficients** β_3, β_4**.** If those were zero, τ would be the same for everyone and `cate()` would collapse to `ate()`. This is why "is the effect heterogeneous?" is literally the question "is β_3 (or β_4) non-zero?" — which we answer with its posterior in Step 5.

Identification. We still need the email->spend link to be unconfounded. Here email was *randomized*, so unconfoundedness holds by design and the required **adjustment set is empty** — the structural model isn't doing confounding-correction, it's giving us a clean way to *query subgroup effects as functionals of the posterior.* `pathmc` reads the graph we declared and confirms that, under that graph, no adjustment is needed — the software checks our drawing, it cannot verify randomization from data:

```
admissible adjustment set(s) for email->spend: [set()]
identifiable: True
```

(An empty adjustment set `[set()]` means "condition on nothing" — correct, because randomization already made email independent of everything. In an *observational* version we'd see the confounders listed here, exactly as in notebook 05.)

1.3 4 · Estimate — fit the structural model

We fit the equation above in one shot; `pathmc` returns a posterior over every coefficient. The `effects_summary()` table below is the direct read-out: each row is a β with its posterior mean and 94% HDI (*Highest Density Interval* — the tightest interval holding 94% of the posterior mass). The two rows to watch are `b_ev` and `b_eg` — the interaction terms. If their intervals sit clearly away from zero, the segments genuinely respond differently; that's the statistical backing for writing segment-specific rules rather than one blanket policy.

```
segment model convergence: max r-hat 1.000 - min ESS 3335 - divergences 0
```

DAG falsification — the check you'd run on real CRM data. Recovery-vs-truth is only available in simulation; on real data you instead test the DAG's *implied* conditional independences against the data — a wrong DAG shows up as an independence the graph predicts but the data violate.

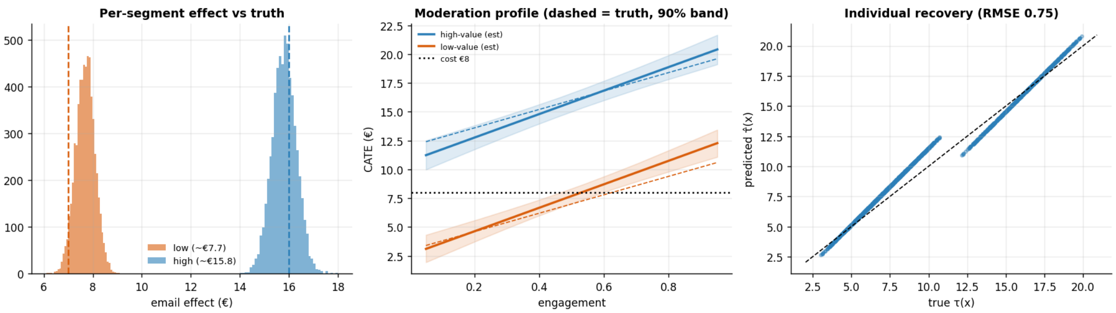
```
DAG falsification: 6 implied independences tested, 0 violated at alpha=0.05
(graph consistent with the data).
```

1.4 5 · Validate — recover the effects, the gap, and the moderation profile

Four checks:

1. **Segment CATEs** vs the planted **€7 / €16** (the low/high segment averages at mean engagement, 0.5) — with credible intervals.
2. **Is the gap real?** the interaction posterior β_3; if it clearly excludes zero, the segments truly differ (not noise).
3. **The continuous moderation profile** — CATE as a smooth function of engagement, recovered vs the true `3 + 9·is_high + 8·engagement` line. This is the payoff of modelling the continuous moderator rather than crudely binning customers into two groups.
4. **Per-customer recovery** — the effect predicted for each individual customer (from the fitted moderation coefficients) against their true effect, on the 45° line. Recovering *segment averages* is easy; recovering the effect for each *individual* is the real test.

```
ATE (pooled)      €10.80   [10.22060982 11.37241506]
CATE low  (eng .5) € 7.69  (true 7.0)   [6.98720232 8.45747794]
CATE high (eng .5) €15.82  (true 16.0)   [14.88273413 16.71767724]
Interaction beta(email:is_high) mean €8.13 (true 9.0), P(>0) = > 0.999 -> gap is
REAL
Interaction beta(email:engagement) mean €10.19 (true 8.0)
```

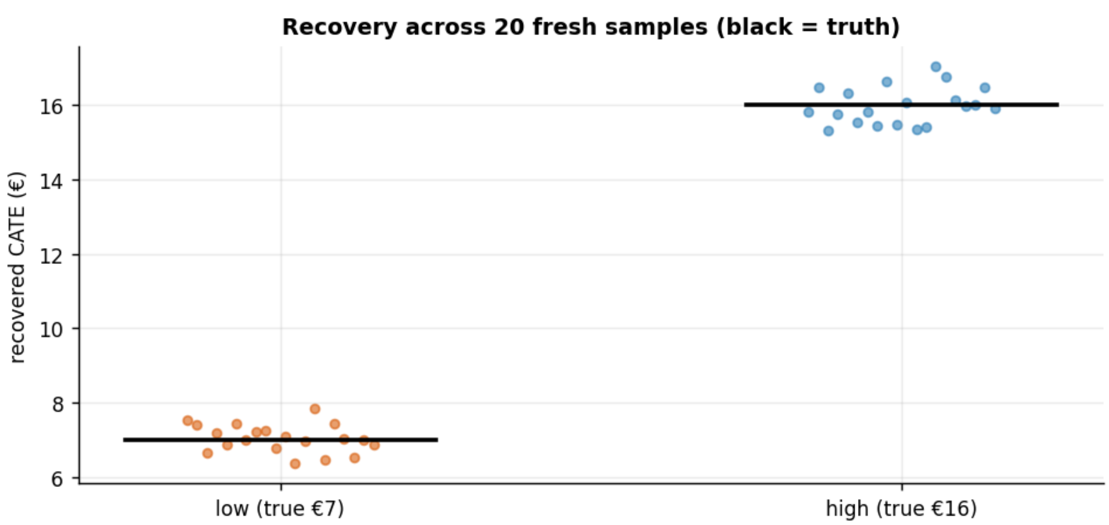


How to read those three panels. *Left* — the two segment-effect posteriors sit right on top of their planted truths (dashed lines) and barely overlap, so the model both **recovers** the effects and **separates** the segments. *Middle* — the estimated CATE (solid) tracks the true `3 + 9·is_high + 8·engagement` line (dashed) across the whole engagement range: modelling the *continuous* moderator paid off, because the effect clearly rises with engagement rather than being a flat per-segment number. Note where each line crosses the €8 cost line — that's the **break-even engagement**, the level above which a customer becomes worth emailing; we quantify it (with a 94% HDI) in the decide step below. *Right* — predicted vs true effect *per customer* hugs the 45° line (low RMSE), confirming recovery at the individual level, not just on segment averages.

1.4.1 5b · Recovery across many seeds — unbiased *and* calibrated?

A single sample can land a little off; the real test of an estimator is whether it is **centred on the truth over repeated samples** (unbiased, not just lucky here) and whether its intervals **cover** the truth at their stated rate. We refit on many fresh samples and check both.

```
low -value: mean €7.05 (true €7)  bias +0.05  sd 0.37  ·  90% HDI covers truth
in 19/20 seeds
high-value: mean €15.98 (true €16)  bias -0.02  sd 0.49  ·  90% HDI covers truth
in 19/20 seeds
```



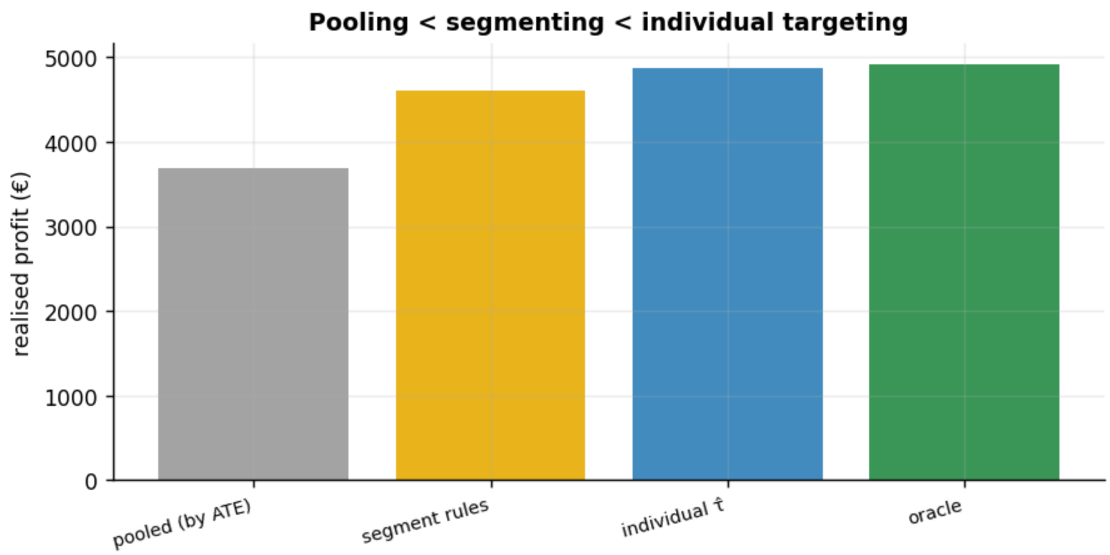
1.5 6 · Decide, in euros — segment rules, and what pooling costs

Now the payoff. First we turn the per-segment effects into concrete **send rules** using the honest probability criterion from notebook 01 — target a segment only where **P(effect > cost) is high** (not merely where the mean beats cost). Then we answer the question that justifies this whole analysis: **how much money does segmenting actually make versus treating everyone as the pooled average?**

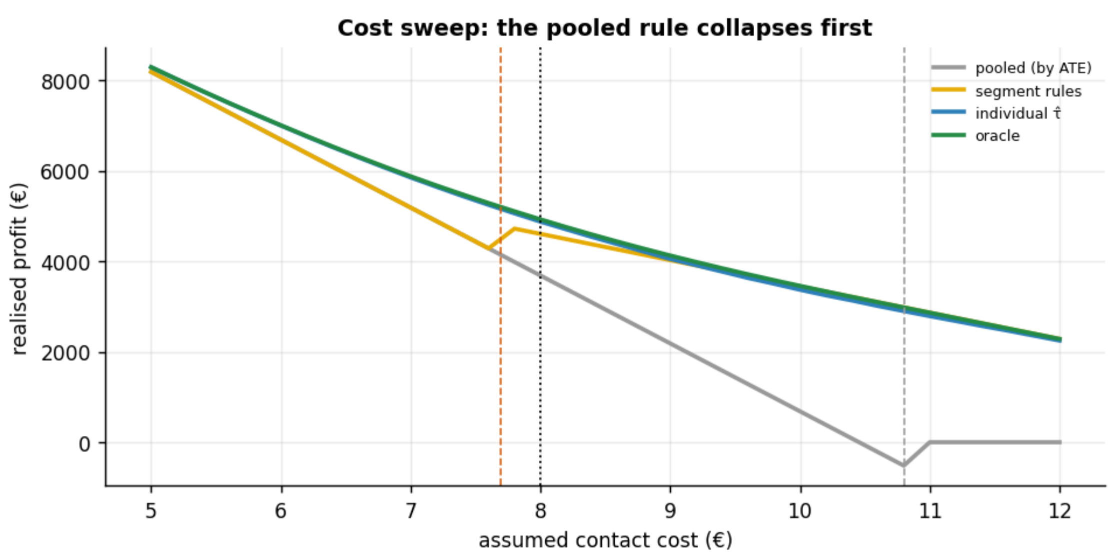
We compare four policies on the *known* per-customer truth (only possible in simulation), forming a ladder of increasing sophistication: **pooled** (one all-or-nothing decision on the ATE) -> **segment rules** (one decision per segment) -> **individual** (a per-customer decision on $\hat{\tau}(x)$) -> **oracle** (the unbeatable rule that knows every true effect). The gaps between the rungs are, literally, the euros that finer targeting is worth.

Two rules, two questions. The four-policy ladder below deliberately uses the risk-neutral posterior-**mean** rule (send whenever *expected* effect beats cost — profit-maximising for expected profit), while the per-segment SEND/TEST/SKIP table above uses the conservative **P(effect > cost) > 0.8** rule; the two answer different questions — how much money finer targeting is *worth* vs. which segments we are *confident enough* to act on now.

```
   segment  effect  net_of_cost  P(effect>cost) action
 low-value    7.69        -0.31            0.21   SKIP
high-value   15.82         7.82            1.00   SEND

Realised profit by policy:
  pooled (by ATE)    €3,682
  segment rules      €4,602
  individual tau_hat      €4,873
  oracle             €4,923

Moving from pooled to individual targeting is worth €1,191 (32% uplift).
```




```
Break-even engagement (where CATE = €8), per segment:
  low-value : g* = 0.53  (true 0.625)   94% HDI [0.46, 0.61]
              -> SEND to low-value customers only ABOVE engagement ~0.53
  high-value: g* = -0.28 (< 0) -> high-value clear €8 at EVERY engagement level
(always SEND)

Cost sweep €5-€12: the segment rule drops low-value once cost passes its CATE
(€7.7), but the
pooled rule treats EVERYONE up to the ATE (€10.8) then flips to treat-NOBODY -
surrendering all the
high-value profit above €10.8, exactly where segment and individual targeting
keep earning.
```



How to read the cost sweep. The four policies earn nearly identical profit only at very low cost, where "email everyone" is optimal. As the contact cost rises they **peel apart at their decision thresholds**: the segment rule stops mailing low-value customers once cost crosses that segment's CATE (€7.7, orange), while the **pooled rule keeps mailing everyone until cost passes the *pooled* ATE (€10.8, grey) — then collapses to mailing no-one**, forfeiting the high-value profit that individual and segment targeting still capture. That brittleness is the cost of one-size-fits-all: a single averaged decision has no way to keep the profitable segment once the *average* stops clearing cost.

The break-even engagement turns the same €8 logic into an operational cutoff: this sample's fitted low-value line crosses €8 at engagement ~**0.53** (94% HDI [0.46, 0.61]) — *mail a low-value customer only once they are more engaged than that.* The true crossing is 0.625; the fitted cutoff sits a touch lower because this sample's low-value CATE runs ~€0.7 high (the small solid-vs-dashed gap in the moderation panel), and a break-even — a **ratio** of two estimated coefficients — amplifies that error near the crossing. That is exactly why you carry the 94% HDI into the decision rather than acting on a single point.

1.6 7 · Caveats

- **Heterogeneity is only as good as the moderator you measured.** If the real driver of differential response isn't `is_high`/`engagement` (or a proxy), the interaction won't catch it.
- **Interaction != causing the moderator.** We're not claiming *making* someone high-value changes responsiveness — only that responsiveness *differs* across pre-existing segments.
- **Random slopes for many thin segments.** With dozens of small segments, replace the single interaction with partially-pooled random slopes so noisy segments borrow strength (see nb 03).
- **Same unconfoundedness caveat** — here email is randomized; observationally a hidden common cause of email and spend would bias every segment effect (see nb 05).